19 October 2026

To the Editors, *General Relativity and Gravitation*

Dear Editors,

We submit our manuscript *"Binary Ellis–Bronnikov wormholes in 3+1 numerical relativity: mergers, fly-bys and gravitational waves"*, by Nikita M. Shirokov and Ilya Nachevsky, for consideration as a Research Article in *General Relativity and Gravitation*.

The manuscript presents, to our knowledge, the first 3D numerical-relativity evolutions of wormhole binaries: pairs of massive Ellis–Bronnikov drainholes held open by a ghost scalar field, evolved with the CCZ4 formulation and adaptive mesh refinement. González, Guzmán and Sarbach established the instability of these wormholes in spherical symmetry (Class. Quantum Grav. **26** 015010 and 015011, 2009), and Kain showed that a collapsed throat settles to Schwarzschild (Class. Quantum Grav. **43** 045014, 2026). We take the problem from one throat in spherical symmetry to two throats in full 3D. The main results are:

- A single throat's e-fold time matches the González–Guzmán–Sarbach linear rate to within 1.9%.
- Two throats obey a charge-sign rule. Like scalar charges repel and opposite charges attract, with a measured attraction-to-repulsion ratio of 1.462 ± 0.022 against the predicted 3/2, so only opposite-signed pairs merge.
- A head-on collision forms a common marginally outer trapped surface around both still-open throats. The horizon then loses 28% of its area as it swallows the phantom field, and by $t = 100$ lies within 1.4% of the Schwarzschild radius of the pair's mass.
- Every encounter radiates a burst of gravitational waves and a scalar dipole of comparable energy and negative sign. A throat survives at most a few orbital periods, and the negative-energy dipole would pump rather than drain the orbit, so these binaries have no inspiral.

The work lies where numerical relativity, exotic compact objects and gravitational-wave sources meet, which we believe suits the journal's readership.

The code, analysis scripts, parameter files and diagnostic streams of every run are public (GitHub tag `v1.0.1-wormhole-merger`, archived at `https://doi.org/10.5281/zenodo.23257999`). A preprint is available as arXiv:2610.13206. The manuscript has not been published and is not under consideration elsewhere, and both authors have approved this submission.

Yours sincerely,

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